

Motorized Controller with Precision Stages

High-Accuracy Motion Control for Scientific & Industrial Systems

Intelligent Optical Imaging.
Automated Microscopy Control.
Built for Research & Inspection.



The CRYONANO Motorized Controller with Precision Stages is a high-precision motion control system designed for applications requiring sub-micron positioning accuracy, repeatable motion, and long-term stability. Built around compact motorized linear and rotary stages with closed-loop electronic control, the system enables smooth, programmable motion for alignment, scanning, and positioning tasks in both laboratory and industrial environments. Its modular architecture allows flexible configuration, making it suitable for integration into optical systems, automated platforms, and precision instrumentation.

KEY FEATURES

- Motion Precision & Performance**
 Sub-micron positioning with high repeatability, smooth low-backlash motion, and closed-loop feedback ensures accurate positioning across multiple cycles.
- Mechanical Design**
 Rigid anodized aluminum and stainless-steel construction with precision crossed-roller bearing stages and lead screw/linear guide mechanisms enables compact, stable operation.
- Motion Control Capabilities**
 Supports step motion, continuous scanning, programmable movement, smooth acceleration/deceleration profiles, multi-axis configurations, and real-time position feedback.
- System Integration**
 Modular single-axis and multi-axis architecture integrates with optical systems, sensors, and probes while supporting external synchronization and automation.

Notes

- Designed for high-precision, low-speed motion control
- Best performance achieved with proper mechanical mounting
- Modular design allows future expansion and upgrades
- Suitable for both standalone and automated systems

APPLICATIONS

- Optical alignment systems
- Precision scanning and positioning
- Semiconductor and microelectronics inspection
- Laboratory automation
- Probe positioning and metrology
- Industrial inspection systems



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SPECIFICATIONS

STAGE SPECIFICATIONS

Stage Type	Linear crossed roller stage
Material	Anodized aluminum / stainless steel
Guiding Mechanism	Crossed roller bearings
Travel Range	25 mm, 50 mm, 100 mm, 200 mm available
Motion Type	Linear / Rotary (modular)

CONTROL & FEEDBACK

Control Mode	Closed-loop motion control
Motion Modes	Step / Continuous / Programmed
Feedback	Real-time position verification

MECHANICAL & SYSTEM

Axes Control	Single-axis / Multi-axis
Interface	USB, Digital I/O
Control Software	GUI with scripting & automation support
Power Supply	24 V DC (standard)
Integration	External control compatible

CONTROLLER SPECIFICATIONS

Axes Control	Single-axis / Multi-axis
Interface	USB, Digital I/O
Control Software	GUI with scripting & automation support
Power Supply	24 V DC (standard)
Integration	External control compatible

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